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Construction Path of Shanghai Oriental Hub Municipal Pipeline Full Cycle Digital Twin Platform

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ABSTRACT

In view of the current status and main problems of the construction of municipal pipelines in Shanghai Oriental Hub, based on the digital twin technology, this paper proposes and constructs a full-cycle digital twin platform for municipal pipelines in the digital era, and discusses the implementation path of the platform construction. On the premise of clarifying the planning and management mechanism of underground pipelines, based on the establishment of a data standard system and safety control scheme, based on building information modeling, IoT perception, artificial intelligence, and other technologies, the deep integration of municipal pipelines and digital twin technology in the whole life cycle of planning, construction, operation and maintenance (O&M) is promoted, which plays an important role in promoting the intensive and efficient construction and safe operation of municipal pipelines in the Oriental Hub. This paper preliminarily discusses the application of digital twin technology in the scenario of Shanghai Oriental Hub Municipal Pipeline Platform, which can provide a reference for 3D pipeline approval, emergency support, and urban physical examination and evaluation.

On February 19, 2024, the State Council officially approved the overall plan for the construction of the Shanghai Oriental Hub International Business Cooperation Zone, marking the official launch of the Shanghai Oriental Hub project. As a world-class large-scale integrated transportation hub, it is an important strategic spatial deployment of a networked urban structure cluster with multiple centers, levels, and nodes [1], Similar to Xiong'an New Area, Shanghai Oriental Hub, as a newly constructed area, has a small historical burden and relatively simple utilization of underground space. The pipeline management system is becoming increasingly perfect, which is conducive to introducing digital twin technology into the construction and application of municipal pipeline platforms. In the Oriental Hub, it gradually realizes the advanced stage of smart city—digital twin city [2]—providing examples and references for other cities to carry out similar work.

The traditional pipeline geographic information platform has realized the integrated development from 2D to 3D and from a single application to multiple technologies, such as the visualization integration platform of VISTA in the United Kingdom [3] and the underground integrated pipe network platform established in major cities in China. This paper proposes a full-cycle digital twin platform for municipal pipelines for the smart Oriental Hub, which is an important part of the overall architecture of the digital twin city, which is a physical space twin, a multidimensional information integration and a people-oriented perception body with the water, electricity, gas, information and other municipal pipelines and comprehensive pipe corridors and their ancillary facilities of the Oriental Hub as the research object. The twin of the physical space is represented by the all-element and refined simulation of the underground pipe network, the

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longitudinal series of the whole process data of the planning, design, construction, operation, and maintenance of the municipal pipeline, and the construction of a three-dimensional and dynamically updated digital base of the municipal pipe network space. The integration of multidimensional information is manifested in the aggregation and sharing of pipeline data from multiple sources, the horizontal breaking of information barriers between government departments and municipal pipeline enterprises, such as planning and design, construction management, and public service operation, and realizing the collection and management of municipal pipeline digital assets such as spatial location, ownership information, operation and maintenance (O&M) status, and emergency response. The people-oriented perception body is manifested as dynamic supervision and iterative optimization of the underground pipe network, real-time monitoring, accurate evaluation, and scientific decision-making, so as to realize the collaborative interaction between the digital space and the physical space of the underground pipe network, promote the reasonable planning, construction and management of municipal pipelines, and ensure the safety and stability of urban operation.

Compared with the traditional pipeline geographic information platform, the full-cycle digital twin platform for municipal pipelines proposed in this paper has a richer extension, focusing on three aspects: policy innovation, management collaboration, and information sharing. In terms of policy innovation, the platform is used as a test field for policy innovation in the Oriental Hub, and it is the first to try, so as to explore the way in advance for rational layout and scientific policy-making; In terms of management coordination, relying on the Oriental Hub platform, we will establish a multi-party collaborative path such as a comprehensive pipeline management department, an industry management department, and a ownership unit, clarify the division of rights and responsibilities, improve the supervision system, and form a municipal pipeline co-construction co-management mechanism. In terms of information sharing, we will strengthen the security control of pipeline information, and build the information sharing service of the Eastern Hub Municipal Pipeline in a hierarchical and classified manner, so as to better play the role of pipeline information in the management of the whole life cycle of infrastructure and optimize the business environment.

1 | Basic Information of Shanghai Oriental Hub

1.1 | Project Overview

The Shanghai Oriental Hub is a "super project" that is related to the implementation of national strategies and the long-term development of Shanghai, aiming to create a world-class large-scale comprehensive transportation hub. The project is located in Zhuqiao Town, Pudong New Area, Shanghai, with a total planned area of 155 km^2, including 84 km^2 of main functional area and 71 km^2 of expansion area. Its core area is centered around Shanghai East Railway Station, connected to Pudong International Airport to the east, forming a transportation pattern of "air rail integration," which will further enhance Shanghai's transportation efficiency and overall competitiveness.

The Oriental Hub is a comprehensive transportation hub created by the strong alliance between Shanghai Pudong International Airport and Shanghai East Railway Station (under construction), integrating air and rail transportation. Its specific location is the area enclosed by the Chuanyang River, the eastern perimeter of Pudong Airport, Xiayan Road, and Pudong Canal. In terms of spatial layout, the plan forms an open and chessboard style overall spatial pattern of "one center, two axes, two rings, four wings, and nine clusters." (Figure 1)

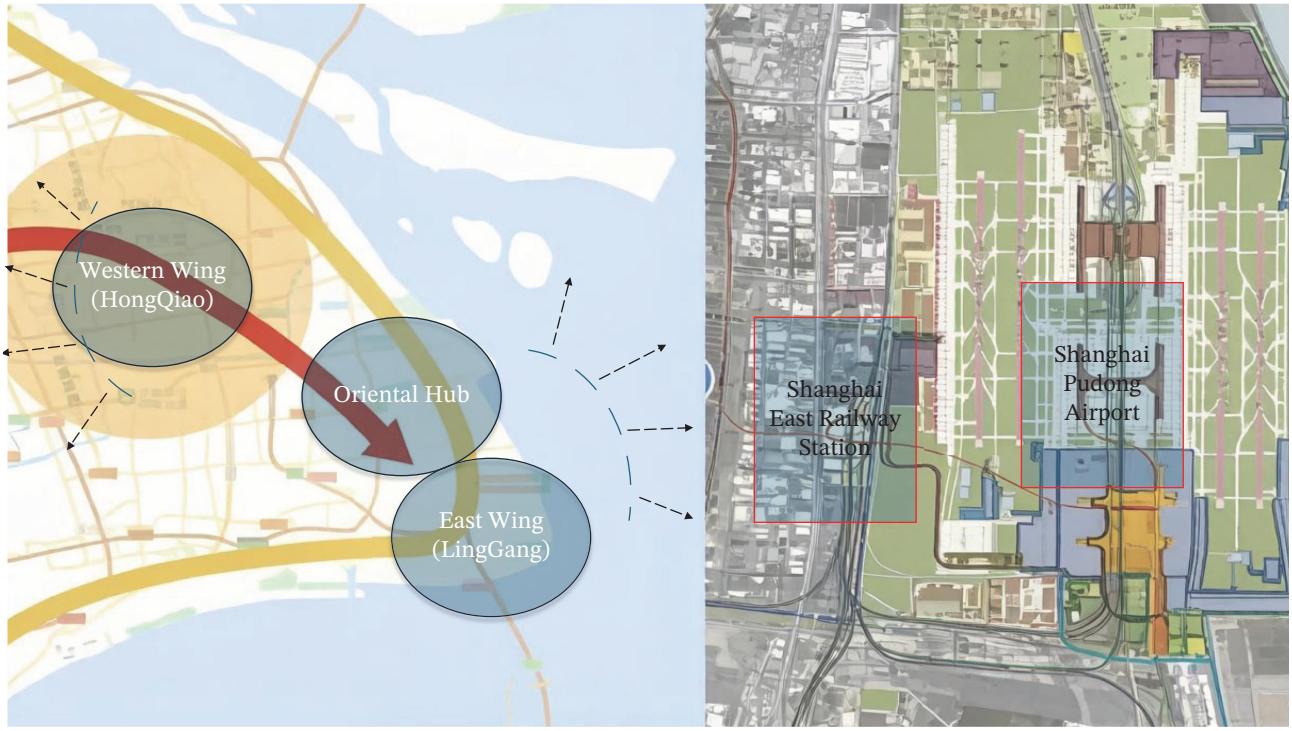


FIGURE 1 | Shanghai Oriental Hub planning map (due to restrictions from the source, this image only uses Chinese standards).

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Among them, "One Heart" is the ecological green heart, creating an ecological green heart that interweaves blue and green, and integrates water and green, with a total area of about 20 km^2. It is the main space for ecological green development in the eastern hub and surrounding areas. The "Two Axes" refer to the open innovation axis that connects the east and west with the river and the sea, and the historical and cultural axis that connects the towns in the north and south.

The "Shuanghuan" is composed of two characteristic functional rings based on eight distinctive major transportation corridors. Among them, the internal functional ring is formed by relying on Liuchen Road (Airport Avenue), Liufeng Road (Star Avenue), Chuannan Feng Road (Aviation Avenue), and Xiayan Road (Zhenxing Avenue); the external functional ring is formed by relying on Jinke Road (Science Avenue), Huaxia Elevated Road (Huaxia Avenue), Hunan Road (Xiwang Avenue), Haitian Avenue, et cetera.

The "Four Wings" are centered around the ecological green heart, with the construction of Oriental Aviation City, Huinan Future City, Zhangjiang Science City, and International Cultural and Tourism City in the southeast, northwest, and northwest respectively. By integrating and enhancing regional spatial patterns, we aim to create a new quality productivity functional zone with global influence.

The "Nine Clusters" are based on the "well" shaped ecological network, highlighting the characteristic functional orientation, forming a total of nine characteristic functional clusters: Ecological Green Heart Cluster, Oriental Hub Cluster, Heqing Suburban Cluster, Laogang Cluster, Huinan Yixuan Bridge Cluster, Xinchang Cluster, Chuansha Yitang Town Cluster, Zhangjiangbei Cluster, and Zhangjiannan South Cluster.

1.2 | Overview of Municipal Pipeline Network for the Project

The municipal pipe network of the Shanghai Oriental Hub is an important part of its overall planning and construction, aiming to ensure the normal operation and efficient management of the hub area. The following is only a list of water supply and energy to illustrate the degree of the municipal pipe network of the Shanghai Oriental Hub.

In terms of water supply, the maximum planned daily water demand of the Eastern Hub is about 43,000 m^3/day. The pressure at the end of the water distribution network is not less than 160 kPa, and the leakage rate of the water supply network is less than 3%. The water supply plan is jointly supplied to the region by Huinan Water Plant, Nanhui North Water Plant, and Yingbin Water Plant. Unconventional water sources, such as collected rainwater, are used for miscellaneous water—such as greening and watering—and municipal tap water is used as a supplementary water source. The water from Huinan Water Plant, Nanhui North Water Plant, and Yingbin Water Plant is transported to Zhuqiao Pumping Station along the G1503 current water pipeline and Zhoudeng Highway planned water pipeline respectively, and the water supply is supplied to the surrounding area after being pressurized by the pumping station. Implemented the Zhuqiao Pumping Station Renovation Project to ensure water supply in the area. Within the planning scope, it is planned to form a "four horizontal and three vertical" water supply backbone pipe network layout, which are DN500 water supply pipe of Wenju Road, DN500 water supply pipe of Zhupan Road, DN500 water supply pipe of Zhuqin Road and DN500–DN600 water supply pipe of Jinting Highway. The "three verticals" are the DN500 water supply pipe of Chuannanfeng Highway, the DN500 and DN600 water supply pipes of G1503 Highway and the DN500 water supply pipe of Zhandong Road. DN300 water supply pipes are laid under other roads, and the water supply pipe network is connected into a ring to meet the water supply safety during peak hours, accidents, and fire fighting. Through DMA partition management, online supervision and other measures, the leakage rate of the pipe network is reduced.

In terms of energy, the energy system of the Eastern Hub focuses on improving the layout of pipeline network facilities. The first is to improve the regional power grid around the 500 kV far East substation in accordance with high reliability standards. The area will build seven new 220 kV substations, renovate four 220 kV substations, and add 18 new 110 kV substations. Two high-sub-high pressure regulator stations and four high-medium pressure regulator stations will be built, and a high-standard high-pressure ring network system will be built to promote the integrated supply of regional gas services and greatly improve the regional power supply and gas supply capacity. The second is to focus on the integrated layout of the energy network. Three 220 kV substations and 12 110 kV substations were newly built, and the pipeline network layout was optimized by using multiple gas sources from West Gas and Sichuan Gas. It is the first in the country to realize the construction of inter-provincial distribution networks of 110 kV and below, explore the establishment of an integrated regional electricity and gas information service platform, and transform from the end of energy facilities to interconnection nodes, so as to greatly improve the reliability of energy supply and service levels.

1.3 | Main Existing Problems

First, the leading role of planning is not enough. At present, the Oriental Hub is playing a leading role in planning [4], designing the top-level structure of the digital transformation of the underground pipe network, and creating a demonstration and leading model area. However, there are still deficiencies in comprehensive consideration of urban road planning, sponge city construction, comprehensive construction of municipal pipelines and pipe corridors, and overall demand promotion, and have not yet been able to establish a working system of planning a map, building a game of chess, and supervising a network. Second, the efficiency of information coordination and sharing needs to be improved. At the planning level, factors, such as the basic conditions, construction difficulty, and landscape requirements of pipeline construction were not considered as a whole. At the data level, problems, such as insufficient co-construction and sharing still exist. Therefore, it is urgent to build a pipeline sharing information platform based on unified data standards, systems and specifications for the whole life cycle of municipal pipelines, promote business integration and open up data flow through the platform, and then realize the data integration and resource sharing of integrated pipelines and professional pipelines. Third, the comprehensive empowerment of digital and intelligent transformation is insufficient. The Eastern Hub has

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initially formed an intelligent and refined urban governance pattern, and some pilot areas have realized the access of water, electricity, gas, and other pipeline data to the existing IoT sensing data such as sponge city and "one network unified management," but the application of new technologies such as big data, cloud computing, and digital twins to realize the aggregation and governance of perception data is insufficient.

1.4 | Comparison With Existing Practices

To highlight the innovative value and scenario adaptability of the proposed platform, this section compares it with typical domestic and international digital twin practices in municipal pipeline management from three core dimensions: technical architecture, application scenarios, and management mechanisms. The comparison clarifies the differentiated advantages of the platform, providing a theoretical basis for subsequent construction ideas and implementation paths (Table 1).

As illustrated in the comparison, existing practices have obvious limitations in adapting to the refined management needs of hub-type pipeline networks: Xiong'an's digital twin focuses on macro urban governance, traditional Shanghai platforms lack full-lifecycle coverage, and international cases are not tailored to "air-rail integration" scenarios. This platform, centered on the Oriental Hub's unique requirements, integrates full-lifecycle data flow and multi-stakeholder collaboration, filling the gap between macro system frameworks and micro pipeline management. Its differentiated design not only ensures the practicality of the project but also provides a replicable paradigm for digital twin applications in similar hub-type municipal pipeline projects .

1.5 | Research Distinctiveness: The "Air-Rail Integration" Context

The "air-rail integration" nature of the Shanghai Oriental Hub necessitates a platform fundamentally different from those for conventional municipal areas. Its distinctiveness stems from addressing three compounded challenges: (1) multimodal data fusion, requiring the integration of pipeline IoT data with dynamic operational schedules from aviation and rail systems

TABLE 1 | Comparison of core dimensions between this platform and typical domestic and international digital twin practices in municipal pipeline management.

Comparison dimension	Typical practices	Core differences of this platform
Technical architecture	Xiong'an New Area's digital twin system adopts a city-wide macro framework, covering all above-ground and underground elements with a "BIM0–BIM5" phase system, focusing on overall urban governance rather than pipeline-specific refinement. Traditional pipeline platforms in other Shanghai regions rely on single-phase applications (e.g., operation and maintenance visualization) with fragmented data flow between phases. International practices (e.g., Queensland's SEQ digital twin) emphasize cross-agency data integration but lack full-lifecycle pipeline data tracing	Focuses on the closed-loop management of pipelines throughout the "planning-design-construction-operation" lifecycle, adopting a "BIM0–BIM6 full-phase connection + lightweight BIM-GIS integration" architecture. It retains micro-level pipeline precision (e.g., LOD500 model accuracy) while realizing real-time data synchronization with IoT, addressing the gap between macro urban frameworks and single-phase local applications
Application scenarios	Xiong'an's system supports multidomain collaboration (transportation, ecology, and municipal services) with pipelines as a secondary component. Shanghai's traditional platforms focus on basic scenarios such as routine inspections and fault reporting, failing to adapt to complex hub environments. International cases (e.g., Barcelona's smart grids) prioritize sustainability over intermodal transportation coordination	Deeply adapted to the "air-rail integration" scenario of Shanghai Oriental Hub, focusing on featured needs such as coordinated planning of airport-railway pipelines, collision early warning for dense pipe networks, and emergency linkage scheduling. It simulates 12+ emergency scenarios (e.g., pipeline leakage) to improve response efficiency, which is not covered by general urban pipeline platforms
Management mechanism	Xiong'an implements a "government-led + departmental division" model with limited enterprise participation in data sharing. Shanghai's traditional platforms suffer from data silos due to fragmented management, with sharing restricted to approval data. Queensland's model emphasizes inter-agency coordination but lacks clear rights and responsibilities for multi-stakeholder collaboration	Establishes a multisubject collaborative mechanism involving planning departments, pipeline enterprises, construction units, and operation teams. Data sharing follows the "classification grading + encrypted processing" principle, with clear authority definitions and a data responsibility list. This mechanism breaks cross-departmental barriers and ensures secure data circulation, complementing the inadequacies of single-led management models

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(e.g., ATM and RTM); (2) cross-domain high-stakes synergy, where pipeline management must be coordinated in real-time with airport, railway, and public safety operations during emergencies; and (3) ultra-high resilience demands, as pipeline failures risk cascading into widespread transportation paralysis. Consequently, this platform is not a generic utility digital twin but a mission-critical decision-support system. Its core architecture—from its unified data engine to its cross-domain simulation tools—is specifically tailored to ensure that pipeline infrastructure reliably supports the uninterrupted and safe operation of the integrated transportation hub. This focus defines the study's unique innovative value.

2 | Construction Ideas for the Full Cycle Digital Twin Platform of Municipal Pipelines in the Oriental Hub

2.1 | Construction Objectives

This paper focuses on the critical window period of the planning, construction, and management of the Oriental Hub, combined with the construction requirements of Shanghai's urban digital transformation and high-quality development, studies and formulates the implementation path of the full-cycle digital twin platform of the Oriental Hub municipal pipeline, and innovates the new concepts, technologies, and models of digital and intelligent management of the municipal pipeline of the Oriental Hub in a way empowered by the platform.

2.2 | Platform Roles and Functional Decomposition Across the Full Lifecycle

To address the core challenges of "information silos" and "disconnected processes" in municipal pipeline management, this platform is designed as a central enabling hub that integrates the planning, design, construction, and O&M stages. Its role and primary functions evolve throughout the project lifecycle, aiming to create a seamless digital thread for data flow and value addition.

During the planning stage (corresponding to BIM1–BIM3), the platform primarily serves as a 3D collaborative planning and simulation verification tool. It tackles pain points like frequent planning clashes and difficulties in assessing layout rationality by integrating multisource data (e.g., geology, road networks, and existing utilities) to create a 3D digital context of the hub area. Its key functions include automated 3D clash detection, utility corridor simulation, and conflict analysis with above-ground structures. This facilitates scheme comparison and optimization, translating planning intent into precise, actionable 3D spatial layouts (BIM3 models), thereby reducing late-stage changes and providing a definitive basis for the design stage.

In the design stage (corresponding to BIM4), the platform transforms into a multidisciplinary collaborative design and automated compliance checking environment. It addresses issues, such as conflicts between different utility designs (water, gas, power, and telecom) and inefficient manual code reviews. Operating on a unified BIM data standard, it provides a collaborative workspace. A core mechanism is its built-in regulatory knowledge database and automated rule-checking engine, which validates designs against mandatory codes. Furthermore, it ensures the detailed design model (BIM4) aligns with the planning-stage BIM3 output, accurately conveying planning intent to construction documents and significantly enhancing design quality and efficiency.

At the construction stage (corresponding to BIM5–BIM6), the platform acts as a digital construction management and as-built verification system. To prevent deviations from design and enable structured management of process data, its central function is real-time comparison between the as-built model (BIM5) and the design model (BIM4), with alerts for significant discrepancies. It also integrates field data on progress, quality, and safety, driving comprehensive digitalization of the construction process. Ultimately, it orchestrates data collection to generate a high-fidelity As-built digital asset (BIM6 model), forming a reliable foundation for subsequent O&M.

For the O&M Stage (corresponding to BIM0), the platform evolves into an IoT-driven smart O&M and decision-support core. Confronting traditional O&M hurdles like unknown asset status and slow emergency response, the platform first establishes the as-built model (BIM6) as the authoritative "As-is Baseline Model (BIM0)." By fusing this model with real-time IoT sensor data (pressure, flow, leak concentration, etc.), it enables comprehensive status monitoring, health assessment, and risk prediction. Its advanced capabilities include emergency scenario simulation and mitigation strategy evaluation based on the digital twin, fundamentally shifting the O&M paradigm from reactive repair to proactive alerting and predictive maintenance.

In summary, by assuming distinct yet coherent core roles across these four stages, the platform establishes a continuous digital thread covering "planning–design–construction–O&M." This effectively fosters the integration of data, processes, and decision-making across phases, providing a unified platform for the refined, full-lifecycle management of the Oriental Hub's municipal pipelines.

2.3 | Construction Framework

In accordance with the overall goal of construction, in view of the current status of the digital construction of the Oriental Hub Municipal Pipeline, fully learn from the advanced concepts and experience of urban development at home and abroad, and build a full-cycle digital twin platform for the Oriental Hub Municipal Pipeline that meets the high-quality development needs of Shanghai's megacity. The framework of platform construction mainly includes four aspects: planning and management mechanism, data standard system, security control mechanism, and overall design of the platform, as shown in Figure 2.

In terms of planning and management mechanism, formulate a supporting management mechanism for municipal pipelines within the scope of responsibility for planning resources. In terms of data standard system, we will develop supporting municipal pipeline planning, design, review, acceptance, review and control index system, and achieve delivery data standards. In terms of security management and control mechanism, formulate pipeline information declassification processing plan and shared service plan. In terms of the overall design of the platform, in accordance with the overall requirements of the internet of

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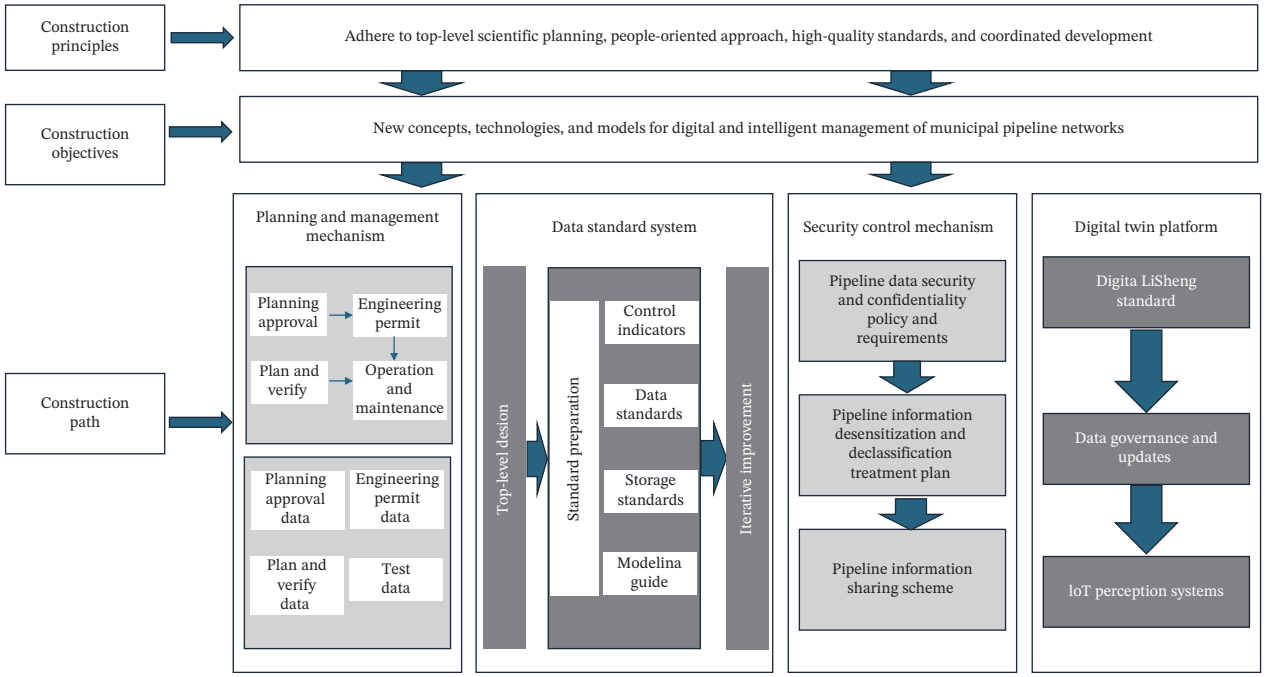


FIGURE 2 | Construction framework of digital twin platform for the full cycle of municipal pipelines in Oriental Hub.

things, the digital connection, and the intelligent connection, the core application scenarios such as intelligent verification of pipeline construction application, pipeline network risk investigation, and emergency auxiliary decision-making are realized.

2.4 | Construction Principles

The construction of the full-cycle digital twin platform for municipal pipelines in the Oriental Hub adheres to four core principles, which are closely aligned with the project's characteristics of "air-rail integration" and the requirements of full-lifecycle refined management. These principles guide the platform's functional design, technical integration, and mechanism construction to ensure practicality, operability, and sustainability.

2.4.1 | Full-Cycle Coverage

The platform ensures continuous coverage of the entire pipeline lifecycle, from planning and design through construction to O&M. This coverage is not merely the sequential inclusion of each stage but is fundamentally about maintaining an unbroken digital thread of information. As detailed in Section 2.2, the platform assumes specific roles at each stage to address stage-specific challenges. Underpinning these roles is the consistent use of 3D BIM models as the core data carrier, which enables the complete preservation, transmission, and traceability of information across phases.

The lifecycle stages, as illustrated in Figure 3, are seamlessly connected through this digital thread: the planning stage (BIM1–BIM3) establishes the foundational layout; the design stage (BIM4) refines the details; the construction stage (BIM5–BIM6) captures as-built conditions; and the operation stage (BIM0) continuously updates the model with real-time status. By maintaining this chain, the platform prevents information fragmentation and ensures that the final built asset aligns with the original planning intent. Moreover, the platform's data management strategy, which will be elaborated in the following subsections, ensures that information is not only passed forward but also can be traced back, supporting accountability and continuous improvement throughout the pipeline's life.

This seamless coverage is technically underpinned by a robust set of data and model management mechanisms, as detailed in the following section.

2.4.2 | Data and Model Management Mechanisms

To ensure the reliability and dynamism of the full lifecycle digital thread, the platform establishes a core set of mechanisms governing model specification, version control, change coordination, and baseline management.

2.4.2.1 | Model Level of Detail (LOD) Grading and Lightweight Flow. The platform defines minimum LOD requirements for models at each stage (e.g., LOD300 for schematic planning and LOD500 for as-built with complete component attributes), following the technical specifications for digital twin of municipal pipelines (DB31/T 1289-2023) [5]. To balance precision with system performance, detailed design and construction models undergo geometry lightweighting and information extraction upon ingestion. This process retains essential spatial topology, key attributes (e.g., diameter, material, and pressure rating), and connectivity. Crucially, a traceability link is maintained to the source model, allowing users to retrieve the full-detail version for specialized engineering analysis when needed.

2.4.2.2 | Data Versioning and Full-Chain Traceability. Structured version control is enforced for core assets (BIM models, critical GIS layers, and approval documents). Each

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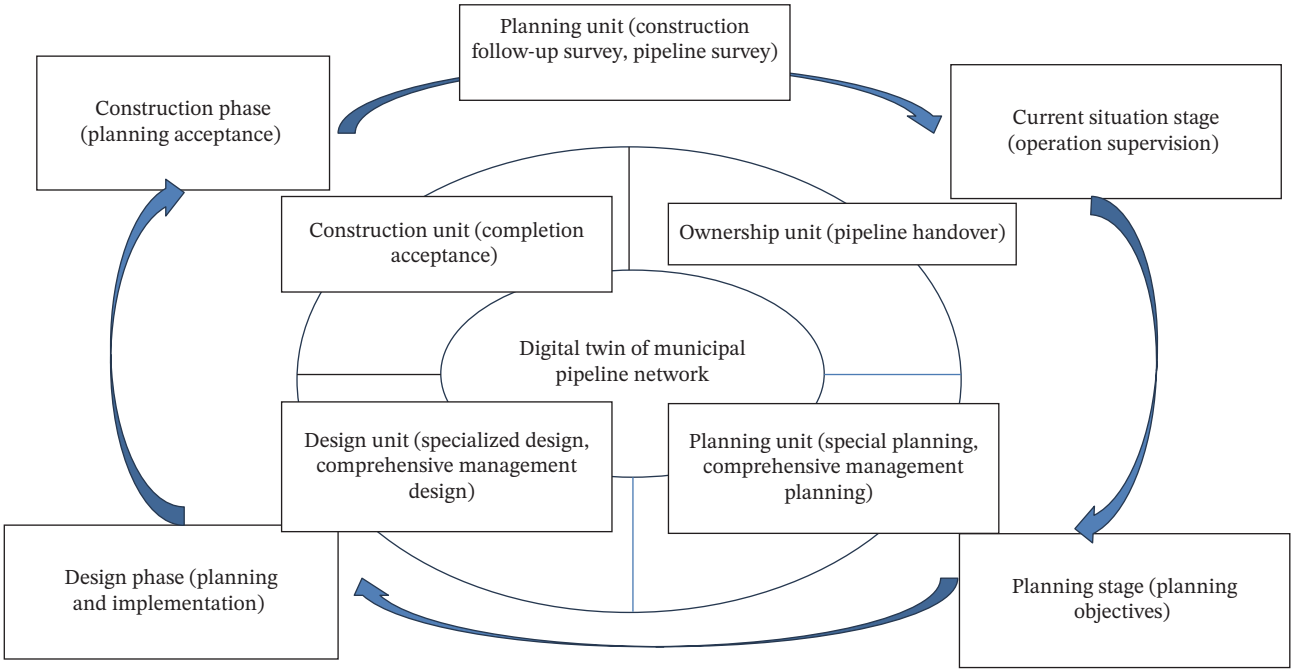


FIGURE 3 | Various stages of the pipeline lifecycle covered by the platform.

submission or significant change generates a new, immutable version tagged with metadata (contributor, timestamp, change description, and approval ID). The platform adopts an incremental versioning schema (e.g., v1.0 and v1.1) and archives all historical versions, preventing data overwriting. This creates a complete "data lineage," enabling historical state tracing and rollback for audit, accountability, and process analysis.

2.4.2.3 | Cross-Stage Change Coordination Workflow. A standardized change coordination process is triggered for design changes or construction adjustments. The workflow includes: initiation and automated impact analysis (leveraging platform clash detection and dependency mapping), online multi-stakeholder review/approval, automated propagation of updates to all affected models/documents, and notification pushes to relevant parties (e.g., downstream contractors and O&M). This workflow ensures that a change approved in the design model (BIM4) is automatically reflected in the ongoing construction model (BIM5) and future maintenance plans, maintaining consistency across stages.

2.4.2.4 | Operational Baseline Establishment and Update Rules. The digital delivery and acceptance process via the platform is critical. The approved as-built model (BIM6) is formally locked and designated as the "Operational Baseline Model" (BIM0), serving as the single source of truth for all subsequent O&M and analysis. Any future physical alteration (e.g., pipeline retrofit) requires the creation of a change order against this baseline. The update process generates a new child version (e.g., BIM0.1) of the baseline after field verification, ensuring the digital twin evolves in lockstep with the physical asset while preserving an accurate change history.

Collectively, these mechanisms ensure that the digital thread is not only continuous but also auditable, controllable, and always representative of the latest physical reality, directly supporting the seamless phase transitions demonstrated in the pilot (Section 4).

2.4.3 | Multi-Party Participation

The platform establishes a multi-stakeholder collaborative mechanism involving urban planning departments, municipal pipeline enterprises (water, electricity, gas, and communication), construction units, and O&M teams. Through permission-based information sharing services, different professional entities can access customized data and functional modules: planning departments obtain integrated planning data for approval; construction units retrieve design and construction standards to avoid rework; and operation teams access real-time monitoring data for daily maintenance. This multi-party participation model breaks the traditional "departmental isolation" and promotes collaborative decision-making throughout the pipeline lifecycle.

2.4.4 | Safety and Controllability

To ensure the secure circulation of sensitive pipeline data (e.g., key hub pipeline layouts and pressure parameters), the platform adopts a "classified grading + decryption processing" mechanism (Figure 4). First, pipeline data is classified into "internal confidential data" (e.g., core hub pipeline coordinates) and "publicly shared data" (e.g., general pipeline layout) based on security levels. Confidential data undergoes desensitization and decryption processing before being shared with authorized entities. Second, a refined permission management system is established to define data access rights for different subjects (refer to Section 2.3.4 for details). Finally, the platform implements full-process data traceability, recording all operations such as data access, modification, and sharing to ensure accountability. This mechanism balances data sharing efficiency and security, complying with national and Shanghai municipal regulations on underground pipeline information management [6].

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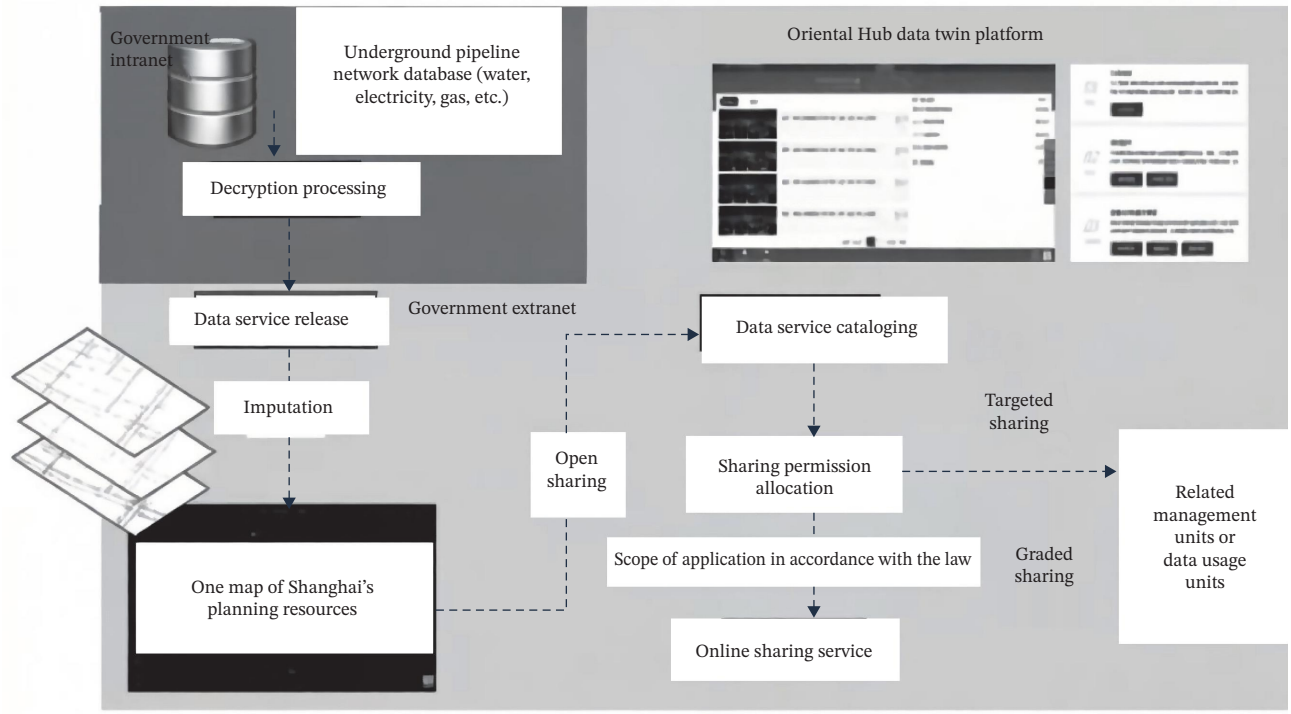


FIGURE 4 | Pipeline classification and shared decryption architecture.

2.4.5 | Compatibility and Extensibility

The platform is designed with strong compatibility to adapt to the existing planning and management systems of the Oriental Hub. Its functional modules (planning, design, construction, and operation) can operate independently according to professional needs or be integrated into a unified management system. In terms of data compatibility, the platform supports the import and export of multiple data formats (e.g., BIM model formats: IFC and RVT; GIS data formats: Shapefile and GeoJSON) to connect with existing urban spatial information platforms. In terms of extensibility, the platform reserves interfaces for new technologies (e.g., AI large models for intelligent decision-making) and new scenarios (e.g., carbon emission monitoring of pipelines) to meet future upgrade needs. Through data expansion and functional iteration, the platform can continuously adapt to the deepening of the Oriental Hub's digital transformation [7].

2.4.6 | Problem-Solution Correlation and Collaborative Mechanism

To address the core problems identified in Section 1.3, the platform establishes a targeted correlation between existing issues, solutions, and implementation paths (Table 2). A cross-departmental data sharing responsibility list (Table 3) is further formulated to clarify the rights and responsibilities of multiple subjects, ensuring the closed-loop management of "problem identification-solution implementation-effect feedback."

The above problem-solution correlation system and responsibility list are deeply integrated with the platform's construction principles. By linking each core problem to specific functional modules and clarifying the data sharing responsibilities of all parties, the platform effectively addresses the disconnection between problems and solutions in traditional management. This collaborative mechanism not only improves the efficiency of pipeline planning, construction, and operation but also provides an institutional guarantee for the optimal allocation of resources in the Oriental Hub, laying a solid foundation for subsequent technical integration and platform construction.

2.5 | Support Technology

To enhance the implementability of the digital twin platform, this section supplements the core technical parameters and operation rules of key technologies such as BIM, IoT sensing, and BIM-GIS integration, combined with industry standards and the characteristics of the Oriental Hub. The specific technical parameters are detailed in the following Table 4.

2.6 | Supplementary Explanation

The above technical parameters are formulated based on the actual pipeline layout of the Oriental Hub (e.g., "four horizontal and three vertical" water supply backbone networks) and the latest industry standards. The BIM phase division and data elements refer to the Shanghai local standard 《Technical Specifications for Digital Twin of Municipal Pipelines》 (DB31/T 1289-2023), ensuring consistency with local construction requirements. The IoT sensing deployment plan is optimized according to the "air-rail integration" scenario characteristics, with encrypted monitoring in core areas to improve the accuracy of risk identification. The BIM-GIS integration adopts a lightweight model to avoid excessive computing resource consumption, which is suitable for the high concurrency requirements of the hub's pipeline management.

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TABLE 2 | Correlation between core existing problems, corresponding platform solutions and implementation paths.

Existing core problems	Corresponding platform solutions	Implementation paths and key measures
Insufficient planning leading role: inadequate integration of urban road planning, sponge city construction, and pipeline corridor construction; lack of a unified "one-map planning, one-game construction, one-network supervision" working system	Full-cycle BIM-GIS integrated 3D planning and verification; unified planning data base	– Integrate road, sponge city, and pipeline corridor data into a unified planning platform based on BIM0–BIM3 models – Implement automated comparison between BIM2 (unit planning) and BIM3 (detailed planning) to ensure consistency with overall planning (alignment deviation ≤ 0.5 m) – Build a "one-map" supervision module for centralized management of planning approval and implementation tracking
Low efficiency of information collaboration and sharing: lack of overall consideration of pipeline construction conditions; insufficient data co-construction and sharing; inter-departmental information barriers	Unified data standard system; hierarchical and classified data sharing with security control	– Formulate standardized data requirements for pipeline planning, design, acceptance, and operation (refer to DB31/T 1289–2023) – Deploy standardized data interfaces to connect government departments, enterprises, and construction units for real-time data circulation – Implement "desensitization + graded permission" for data sharing to balance openness and security
Inadequate digital and intelligent empowerment: insufficient application of new technologies; lack of perception data aggregation and governance	Integration of IoT sensing, AI, and 5G technologies; intelligent perception computing and emergency early warning	– Access multimodal IoT sensing data (water pressure, gas leakage concentration, pipeline temperature) with a transmission delay ≤ 1 s – Build an AI-based risk identification model to realize automatic aggregation and analysis of perception data – Develop a digital twin-based emergency decision-making module to simulate 12+ accident scenarios (e.g., pipeline burst, gas leakage) to support rapid emergency disposal

TABLE 3 | Cross-departmental data sharing responsibility list for municipal pipeline digital twin platform.

Responsible subjects	Data sharing content	Authority level	Submission requirements
Urban planning and resources department	Pipeline planning approval data, spatial layout data, planning control indicators	Full open access to authorized platform users	Submit in standardized BIM/GIS format within 3 working days after approval
Municipal pipeline enterprises	Operation and maintenance data, equipment status records, fault/leakage reports	Directed open access to supervision departments and collaborative units	Real-time synchronization of key operation data; monthly submission of statistical reports
Construction units	Construction progress data, completion acceptance documents, BIM6 models	Limited open access to planning departments and owner units	Submit BIM6 models and acceptance data within 7 working days after project completion
Operation and maintenance units	Daily inspection records, emergency disposal data, renovation records	Full open access to pipeline enterprises and supervision departments	Real-time update of inspection data; quarterly submission of pipeline health reports

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TABLE 4 | Core technical parameters and operation rules of key technologies for the municipal pipeline full-cycle digital twin platform.

Technology category	Technical details	Core parameters and operation rules	References
BIM technology (BIM0–BIM6 phases)	Covers the full lifecycle of pipelines, with each phase corresponding to different engineering stages. Data elements include spatial location, attribute information, construction records, and acceptance indicators	– BIM0 (current status model): core data elements—pipeline diameter (DN300–DN600), material (steel/PVC), and spatial coordinates (WGS84 coordinate system). – BIM1–BIM3 (planning phases): BIM3 (detailed planning) vs. BIM2 (unit planning) comparison criteria: pipeline alignment deviation ≤ 0.5 m, node spacing consistency ≥ 95% (compliant with DB31/T 1289–2023) – BIM4–BIM6 (design–construction phases): BIM6 (completion acceptance) verifies against BIM3, with acceptance indicators including pipeline pressure resistance (≥1.6 MPa for water supply pipes) and leakage rate (<3%) – Data management: incremental storage is adopted; late-stage data is associated with early-stage models without overwriting, ensuring full-lifecycle traceability	[8, 9]
IoT sensing system	Deployed for real-time monitoring of pipeline operation status, covering water supply, gas, and power pipelines. Multimodal data includes physical parameters, environmental conditions, and alarm signals	– Deployment density: 1 monitoring point per 500 m for water/gas main pipelines; 1 monitoring point per 300 m in core hub areas (e.g., around Shanghai East Railway Station) – Key indicators: water supply pressure (160–200 kPa), gas leakage concentration (<0.1%), data transmission delay ≤ 1 s – Edge nodes deployed in eight pipeline intersection hubs (e.g., Zhuqiao Pumping Station), supporting real-time data preprocessing and local early warning	[6, 10]
BIM-GIS integration	Realizes the conversion of BIM model data to GIS topological structure, supporting spatial analysis functions while retaining BIM visualization advantages	– Data conversion: BIM model exports pipeline topological data (node connections, diameter, material) and automatically adapts to GIS spatial databases – Supported functions: pipeline burst range calculation, connectivity verification, and 3D collision analysis with underground structures (e.g., integrated pipe corridors) – Synergy logic: BIM provides micro-level precision; GIS supports macro spatial analysis, achieving lightweight and efficient collaboration	[11, 12]

3 | The Construction Content of the Full-Cycle Digital Twin Platform of the Municipal Pipeline of the Oriental Hub

Based on the "one map" of territorial spatial planning, under the dual constraints of standards and specifications, information security system, organization management, and operation guarantee system, the top-level design of the infrastructure layer, support engine layer, twin base layer, and intelligent application layer is improved according to the overall architecture, so as to create a future-oriented digital twin platform for the municipal pipeline of the Oriental Hub. The construction of the platform includes four aspects: spatial resource aggregation, 3D spatio-temporal analysis, intelligent perception computing, and digital twin support, and the relevant architecture is shown in Figure 5.

The first is to realize multisource model integration and pipeline thematic governance. On the one hand, on the basis of the existing urban space basic information platform, the three-dimensional data base of the whole space is supplemented, the current model data of municipal pipelines, holographic roads, existing buildings, digital terrain and other current models are

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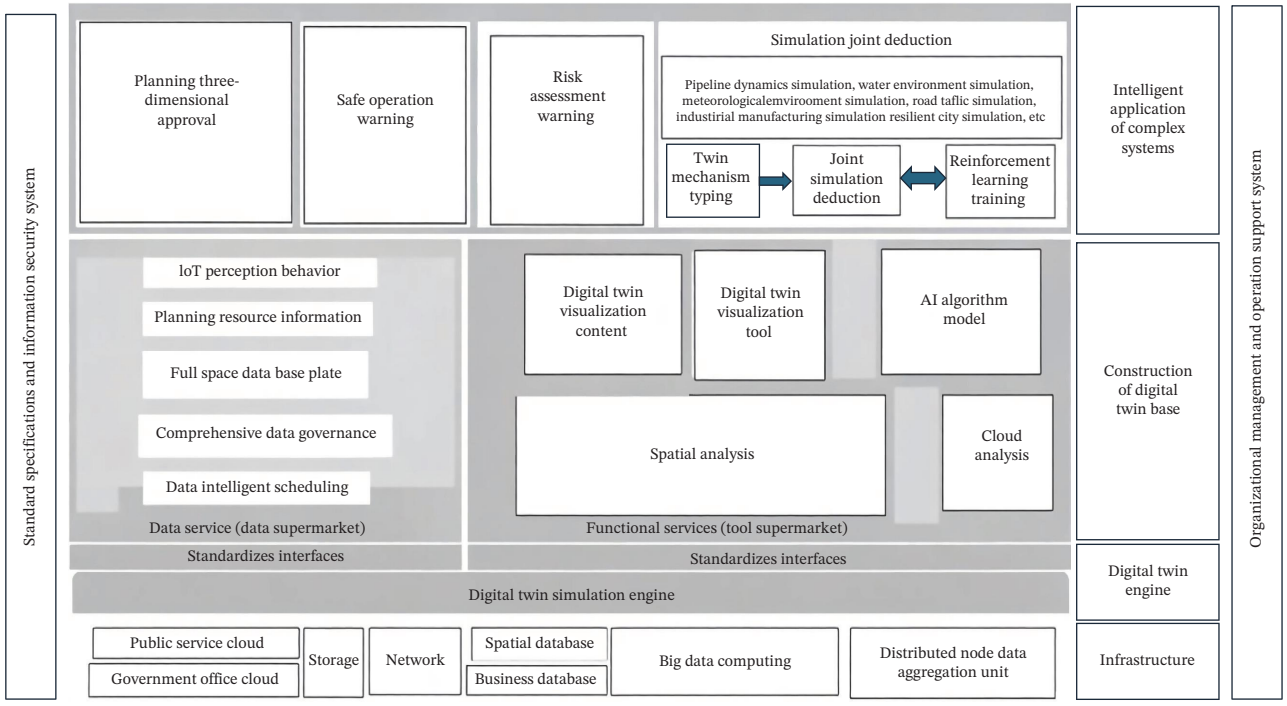


FIGURE 5 | Overall architecture of the digital twin platform for municipal pipelines at the Oriental Hub.

integrated, and the big data of IoT perception and spatiotemporal behavior is accessed through distributed edge nodes, so as to realize the effective connection of spatial data base and planning resource business information in the form of standardized data service interfaces. On the other hand, around the whole life cycle of the pipeline, special governance such as data quality, security, sharing, and standardization is carried out to improve the quality, security, and compliance of pipeline data, and provide guarantee for the effective use of data.

The second is to carry out the construction of three-dimensional analysis capacity of underground space planning. On the basis of the existing two-dimensional spatial analysis capabilities, enrich the "tool supermarket," supplement the three-dimensional simulation and underground space analysis capabilities, and further improve the service level. In terms of digital twin visualization, it supports multisource data and service access, realizes custom assembly, editing and export of 2D and 3D scenes, manages spatial scenes through visual ledgers, and monitors data assets. In terms of 3D analysis capabilities, based on the application needs of relevant government departments, enterprises and institutions, we will build spatial computing tools that support secondary development, including general functions such as pipeline section analysis, and support business applications such as digital approval of 3D pipelines, comparison and selection of design schemes, and collision analysis with surrounding structures, requisition and demolition, relocation and modification simulation, as shown in Figure 6.

The third is to establish a network emergency knowledge graph under the IoT perception system [13]. Build a multimodal AI algorithm and computing power support platform for pipe network risk identification, as shown in Figure 7. From the aspects of algorithm model and infrastructure, it is necessary to consolidate the perceptible and computable capability foundation of the urban digital twin base. Through the pipeline network monitoring system, internet of things sensors, satellite remote sensing, et cetera, the multimodal data of pipeline network operation, including pipeline physical parameters, operation status, surrounding environment, voice alarm, monitoring video, et cetera, is obtained, and a dataset can be used for AI algorithm training through data cleaning, data annotation, and data fusion. According to the specific needs of pipeline network risk identification, the appropriate AI algorithm is selected, and the infrastructure layer provides storage, network, computing nodes, and cloud resources, and builds a knowledge graph of risk management and control and intelligent early warning based on the intelligent perception system of the municipal pipe network, so as to improve the comprehensive resilience of urban pipe networks such as communications, energy, and water supply [13].

The fourth is to explore the intelligent scheduling and cloud management of the city's digital twin engine [14]. Open up the data channel between the GIS engine and the digital twin engine, and intelligently dispatch the data center of the digital twin engine to extract data in real time for analysis and calculation, supporting high concurrency and second-level response. Build the open co-simulation capability of the digital twin platform to realize the dynamic allocation of computing power, service call monitoring, and cloud O&M management. According to the running status and requirements of the pipe network, a combination of rules-based, machine learning-based, and hybrid intelligence-based scheduling rules is used to intelligently allocate the computing resources of the digital twin engine [14]. At the same time, it can monitor the frequency of service calls and the real-time occupation of resources online, realize cloud management and O&M, detect abnormal events in time, and ensure the safe and efficient operation of the pipe network.

Beyond these core technical components, the platform is also designed to deliver advanced, scenario-specific intelligent applications. A prime focus is addressing the unique complexities of the

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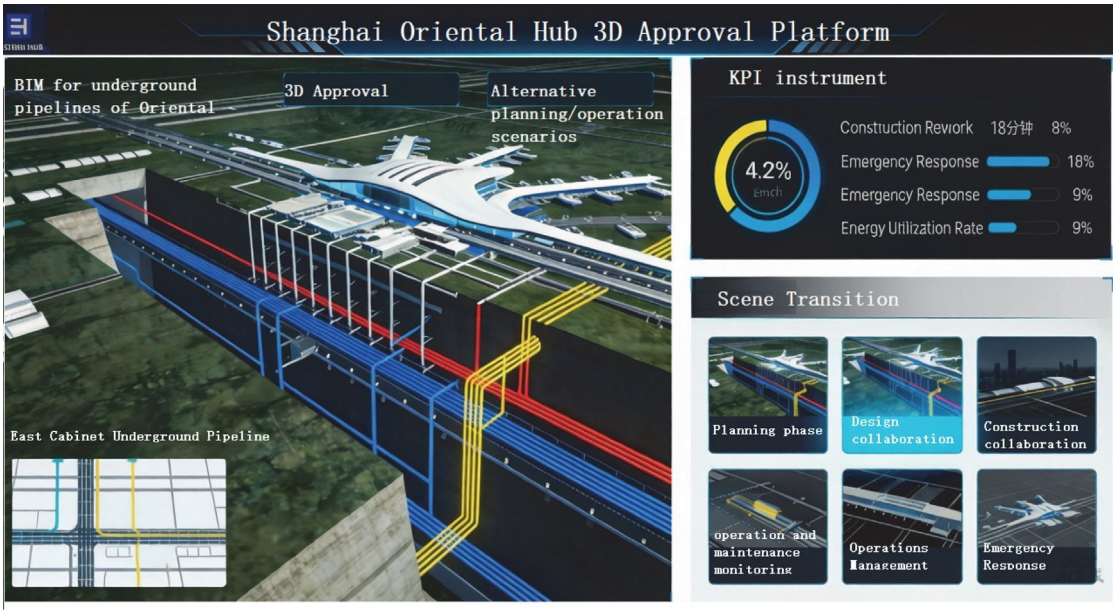


FIGURE 6 | Main functions of 3D analysis of underground space on the platform.

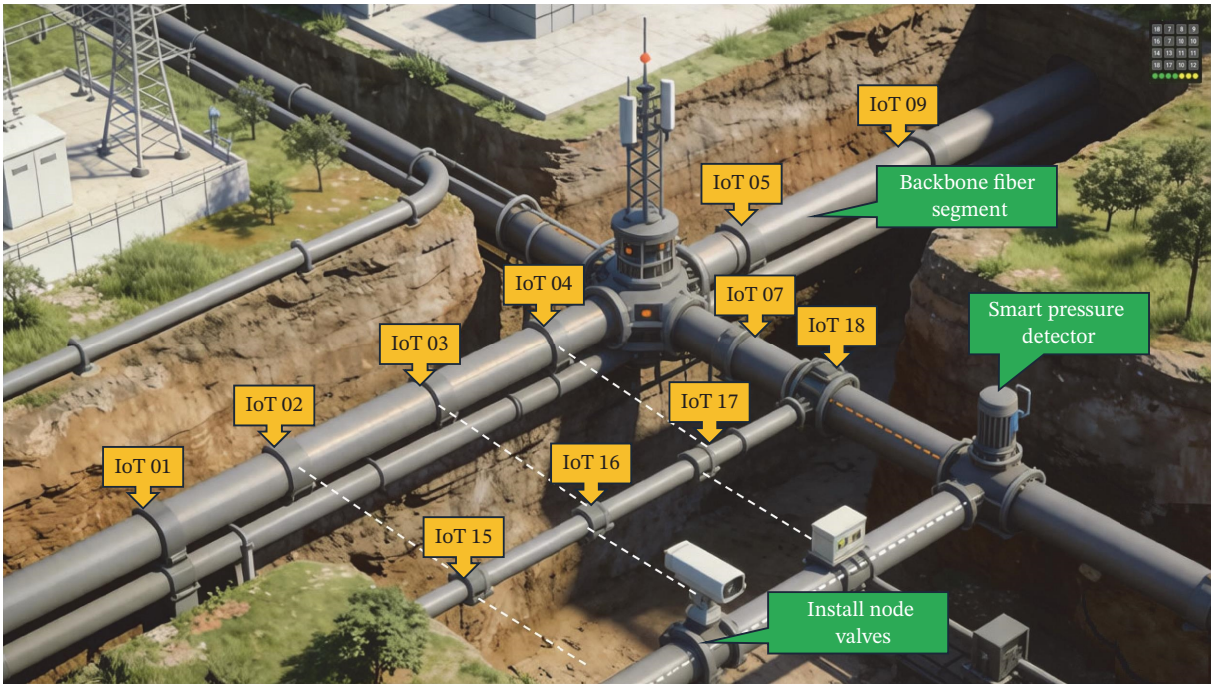


FIGURE 7 | Municipal pipe network platform based on IoT perception system.

"air-rail integration" hub. For instance, by integrating the foundational capabilities of spatial–temporal data aggregation and IoT perception, the platform can perform dynamic utility demand simulation. It correlates real-time flight and train schedules with hydraulic and electrical models of the pipeline network to forecast short-term load peaks, enabling proactive dispatch of resources like pumping stations to transition from static to demand-responsive operations. Furthermore, by leveraging the 3D analysis and digital twin simulation engine, the platform enables integrated cross-domain emergency response. In the event of a pipeline incident, it fuses sensor alarms with real-time passenger flow data within the unified 3D scene to simulate hazard dispersion, automatically generating synchronized response protocols that coordinate utility shutdowns, passenger guidance, and emergency logistics. These targeted applications demonstrate how the platform's generic technical architecture is specifically orchestrated to solve the high-stakes, interconnected challenges characteristic of a world-class transportation hub.

4 | Pilot Application Effectiveness

To verify the practical value and technical operability of the proposed full-cycle digital twin platform, a targeted pilot application was carried out in the core functional area of the Shanghai Oriental Hub (focusing on the 3.5 km^2 area around Shanghai

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East Railway Station, where water supply and gas pipelines are densely distributed). The pilot period lasted 5 months (February 2024–June 2024), covering the planning optimization, construction supervision, and operation monitoring stages of municipal pipelines. Key performance indicators (KPIs) before and after platform deployment were compared to quantify the application effect, with specific details as follows.

4.1 | Pilot Background and Scope

The pilot area is a typical "air-rail integration" node, adjacent to Shanghai East Railway Station (under construction) and 2 km away from Pudong International Airport Terminal 4. The underground pipeline network in this area includes 4 DN500–DN600 water supply main pipes, three DN400 gas pipelines, and two communication cable ducts, with complex spatial intersections (overlapping with two integrated pipe corridors). During the pilot, 28 IoT sensing nodes were deployed (1 node per 300 m in core areas and 1 node per 500 m in general areas) to collect real-time data, such as water supply pressure, gas leakage concentration, and pipeline surface temperature, with data transmission delay controlled within 1 s. The platform's core modules—3D planning verification, BIM-based construction alignment, and IoT-driven operation monitoring—were fully applied to test their adaptability to hub scenarios.

4.2 | Pilot Implementation and Key Results

The pilot focused on solving three core problems in the pilot area: planning collision risks, construction rework caused by information asymmetry, and delayed leakage detection. The specific implementation process and effect data are shown in Table 5.

4.3 | Pilot Experience and Limitations

4.3.1 | Key Experience

The pilot confirms that the platform's core designs are effective for hub-type pipeline management:

The "BIM0–BIM6 full-phase connection" mechanism ensures the consistency of planning intentions and on-site implementation. For example, the BIM3 detailed planning model was directly used as the benchmark for BIM6 completion acceptance, avoiding the "planning-construction disconnection" common in traditional projects.

The "multisubject data sharing" mechanism (based on Table 4) realizes real-time data circulation among the planning department, construction unit, and water supply enterprise. During the pilot, the construction unit accessed the latest planning adjustments through the platform within 1 h, compared with 3 days in the traditional mode.

The integration of IoT and digital twin technology transforms O&M from "passive response" to "active prevention." The platform's AI risk model successfully predicted 1 valve aging risk 7 days in advance, which was verified by on-site inspection.

4.3.2 | Current Limitations

The pilot also exposed two areas for improvement:

Multi-pipeline adaptability: the current platform focuses on water supply and gas pipelines, but lacks dedicated monitoring modules for power and communication pipelines (e.g., cable temperature rise warning). Subsequent optimization needs to supplement IoT sensing parameters for special pipelines.

Cross-regional data connection: the pilot area's pipeline data is not yet connected to Pudong International Airport's pipeline system, making it difficult to support cross-node emergency linkage. Future research needs to explore cross-departmental data interface standards.

4.4 | Pilot Significance

The pilot results show that the platform can effectively improve the efficiency of pipeline planning, construction, and O&M in hub areas, and its core technologies (such as BIM phase management and IoT lightweight deployment) are highly replicable. For example, the BIM-GIS collision analysis method used in the pilot has been promoted to the Oriental Hub's Huinan Future City area, providing a practical reference for the platform's large-scale application. At the same time, the pilot data (e.g., 36.4% O&M cost reduction) also provides quantitative support for the platform's role in promoting the Oriental Hub's "intensive construction and safe operation," laying a foundation for subsequent policy promotion.

5 | Conclusion and Prospect

5.1 | Conclusion

This study focuses on the full-cycle management demands of municipal pipelines in the Shanghai Oriental Hub, a national-level comprehensive transportation hub, and systematically constructs a digital twin platform integrating planning, construction, operation, and maintenance. By addressing the core challenges identified in the project—insufficient planning guidance, inefficient information collaboration, and inadequate digital empowerment—the research achieves three key outcomes:

First, it clarifies the differentiated positioning of the platform. Through comparative analysis with domestic practices (e.g., Xiong'an New Area's macro digital twin city) and international cases (e.g., Queensland's SEQ Digital Twin), this platform is confirmed to fill the gap between "city-level overall governance" and "single-phase pipeline management." Its "BIM0–BIM6 full-phase connection + lightweight BIM-GIS integration" architecture and "air-rail integration" scenario adaptability (e.g., coordinated planning of airport-railway pipelines, dense network collision early warning) provide a targeted solution for hub-type pipeline refined management.

Second, it improves the implementable technical system of the platform. By supplementing core technical parameters (Table 2)—such as BIM phase data elements (DN300–DN600 pipeline diameter and WGS84 coordinate system), IoT sensing deployment standards (1 monitoring point per 300 m in core areas, ≤1 s data delay), and BIM-GIS synergy logic—the research ensures the platform's technical feasibility. These parameters, aligned with Shanghai's local standard DB31/T 1289-2023 and national code GB 50268-2022, lay a solid foundation for on-site application.

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TABLE 5 | Comparison of key performance indicators (KPIs) before and after platform deployment in the pilot area.

Performance indicators	Before platform deployment (traditional mode)	After platform deployment (digital twin mode)	Key supporting platform functions/ mechanisms	Improvement effect
Planning collision detection rate	68% (manual review of 2D drawings, missing 32% of potential risks)	97% (automated BIM-GIS collision analysis, covering pipeline-structure, pipeline–pipeline intersections)	Automated 3D clash detection and spatial coordination analysis based on integrated BIM-GIS models	Increased by 29 percentage points; 4 potential collisions (e.g., water pipe vs. rail foundation) identified and optimized in advance
Construction rework rate	15% (inconsistent on-site construction with planning drawings, e.g., pipeline elevation deviation)	4.2% (real-time alignment of BIM5 construction model with BIM3 planning model, on-site feedback within 10 min)	Real-time virtual-physical comparison and deviation alert using BIM5 site models synchronized with the as-planned BIM3 baseline	Reduced by 10.8 percentage points; construction period shortened by 12 days (from 45 days to 33 days)
Pipeline leakage detection and positioning time	150 min (manual patrol + pressure gauge inspection, positioning error ± 15 m)	18 min (IoT pressure sensor + digital twin simulation, positioning error ±3 m)	IoT-data fusion and hydraulic model simulation within the digital twin for rapid anomaly localization	Shortened by 132 min; 2 minor water leakage incidents handled in time, avoiding water waste of 280 m^3
Emergency response time (gas leakage)	85 min (on-site investigation → scheme discussion → personnel dispatch)	25 min (digital twin scenario simulation → optimal path planning → one-click work order)	Scenario-based emergency simulation and one-click dispatch of optimized mitigation plans	Shortened by 60 min; impact range of simulated gas leakage reduced by 45% (from 200 to 110 m^2)
Monthly operation and maintenance (O&M) cost	198,000 RMB (including 12 manual patrol personnel, regular equipment inspection)	126,000 RMB (intelligent monitoring + predictive maintenance, reducing 6 patrol personnel)	Predictive maintenance alerts and optimized resource dispatch driven by IoT analytics and asset health models	Reduced by 36.4%; predictive maintenance of three aging valves avoided unplanned shutdowns

Note: The platform's scenario library is designed around a risk-based taxonomy, covering four primary categories. (A) External interference (e.g., third-party excavation damage and vehicle impact). (B) Material and corrosion failure (e.g., pipe weld failure, internal corrosion leakage, and joint seal failure). (C) Geotechnical and hydraulic issues (e.g., ground settlement inducing pipe stress, water hammer, and scour erosion). (D) Operational and equipment faults (e.g., valve/pump failure, control system malfunction, and over-pressure events). For simulation, key parameters are defined. For instance, a gas pipeline leakage scenario may be configured with a leak orifice size (e.g., 10 mm diameter) and environmental conditions (e.g., wind speed of 2 m/s and stable atmosphere). A water main burst scenario may specify burst type (longitudinal split), initial flow rate, and surrounding soil permeability to model flooding extent.

Third, it establishes a closed-loop problem-solution mechanism. Through the "core problems-solutions-implementation paths" correlation system (Table 3) and cross-departmental data sharing responsibility list (Table 4), the platform effectively breaks inter-departmental information barriers. For example, the unified planning database (integrating road, sponge city, and pipeline corridor data) strengthens planning guidance, while standardized data interfaces realize real-time collaboration among 4+ stakeholder groups (planning departments, pipeline enterprises, and construction units). This mechanism directly improves the efficiency of pipeline planning approval and reduces construction rework risks.

Practically, the platform contributes to the Oriental Hub's development in three aspects: promoting intensive construction of water supply, gas, and power pipelines; optimizing resource allocation among pipeline ownership units; and enhancing accident response capabilities. It also provides a replicable paradigm for digital twin applications in other comprehensive transportation hubs (e.g., Beijing Daxing International Airport and Guangzhou South Railway Hub).

6 | Prospect

While this research realizes the basic framework of the full-cycle digital twin platform, future studies can be expanded in three directions.

6.1 | Multi-Type Pipeline Adaptability Optimization

The current research focuses on water supply and gas pipelines. Subsequent work will extend the platform's functions to power and communication pipelines, supplementing IoT sensing parameters for special pipelines (e.g., temperature monitoring for power cables and signal strength detection for communication pipelines) and optimizing the AI risk identification model to

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cover multi-pipeline coupling fault scenarios (e.g., gas pipeline leakage affecting adjacent power cables).

6.2 | Deep Integration of Emerging Technologies

Explore the application of "digital twin + AI large models" to realize intelligent generation of pipeline maintenance plans and automatic optimization of emergency disposal schemes. For example, based on 3-year+ operation data, the model can predict pipeline aging trends and proactively push maintenance reminders, transforming management from "passive response" to "active prevention."

6.3 | Cross-Regional Collaboration Mechanism Expansion

With the Oriental Hub's inter-provincial radiation role, future research will explore the connection of the platform with pipeline digital twin systems in neighboring cities (e.g., Suzhou, Ningbo). By formulating unified cross-regional data sharing standards and emergency linkage protocols, it will build a regional integrated pipeline safety guarantee system, supporting the coordinated development of the Yangtze River Delta urban agglomeration.

In summary, the full-cycle digital twin platform for municipal pipelines is not only a key support for the Oriental Hub's digital transformation but also an important practice for promoting the intelligent upgrading of urban infrastructure. With continuous optimization of technology and mechanisms, it will play a more significant role in ensuring urban operation safety and improving resource utilization efficiency.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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